

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name : F2 GP Thinner

Other names/ synonyms: N/A

Recommended use/ Restrictions on use: As an organic thinner

Supplier: LSC CHEMICAL & PAINTS SDN. BHD.

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2. HAZARDS IDENTIFICATION

GHS Classification: Flammable Liquids, Category 2
 Acute Toxicity (Oral), Category 3
 Acute Toxicity (Dermal), Category 3
 Acute Toxicity (Inhalation), Category 3
 Skin Corrosion/ Irritation, Category 2
 Serious Eye Damage/Irritation, Category 2
 Carcinogenicity, Category 2
 Toxic to Reproduction, Category 1A
 Specific Target Organ Systematic Toxicity (Single Exposure)
 - Central Nervous System, Eye, Respiratory Tract, Category 1
 - Narcotic effects, Category 3
 Specific Target Organ Systematic Toxicity (Repeated Exposure)
 - Central Nervous System, Kidney, Liver, Category 1
 Aspiration Hazard, Category 1

GHS Label Element/ Pictogram:

Signal Words: Danger

GHS Hazard Statement:
Physical Hazards: H225: Highly Flammable liquid and vapor.

Health Hazards : H301: Toxic if swallowed.

H304: May be fatal if swallowed and enters airways.

H311: Toxic if in contact with skin.

H315: Causes skin irritation.

H319: Causes serious eye irritation.

H331: Toxic if inhaled.

H351: Suspected of causing cancer.

H336: May cause drowsiness or dizziness.

H360: May damage fertility or the unborn child.

H370: Causes damage to organs (CNS)

H372: Causes damage to organs (CNS, kidney, liver) through prolonged or repeated exposure.

Environmental Hazards: Not classified as environmental hazards under GHS criteria.

**GHS Precautionary Statements:**

- Prevention** : P201: Obtain special instructions before use.
P202: Do not handle until all safety precautions have been read and understood.
P210: Keep away from heat/sparks/open flame/hot surface– No smoking.
P240: Ground/bond container and receiving equipment.
P241: Use explosion-proof electrical/ventilating/lighting equipment.
P242: Use only non-sparking tools.
P243: Take precautionary measures against static discharge.
P260: Do not breathe fume/mist/vapors/spray.
P264: Wash hands thoroughly after handling.
P270: Do not eat, drink or smoke when using this product.
P271: Use only outdoors or in a well-ventilated area.
P280: Wear protective gloves/protective clothing/eye protection/face protection.
P281: Use person protective equipment as required.
- Respond** : P301+P310: IF SWALLOWED: Immediately call a POISON CENTER or doctor/ physician.
P302+P352: IF ON SKIN: Wash with plenty of soap and water.
P303+P361+P353: IF ON SKIN (or hair): Remove/take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P307+P311: IF exposed: Call a POISON CENTER or doctor/ physician.
P308+P313: IF exposed or concerned: Get medical advice/ attention.
P332+P313: If skin irritation occurs: Get medical advice/ attention.
P337+P313: If eye irritation persists: Get medical advice/attention.
P370+P378: In case of fire: Use appropriate media for extinction.
P312: Call a Poison center or doctor/physician if you feel unwell.
P314: Get medical advice/ attention if you feel unwell.
P331: Do not induce vomiting.
P362: Take off contaminated clothing and wash before reuse.
- Storage** : P403+P235: Store in a well-ventilated place. Keep cool.
P233: Keep container tightly closed.
P405: Store locked up.
- Disposal** : P501: Dispose of contents and container to appropriate waste side or reclaimer in accordance with local and national regulations.

Other Hazards Which Do Not Result In Classification:

Not available

3. COMPOSITION/INFORMATION ON INGREDIENTS**Product/ Chemical Identity:** F2 GP Thinner**Chemical Characterization:** Mixture

Chemical Name	CAS Number	Concentration
Mixture of aromatic hydrocarbon	-	30%-60%
Alcohol	67-56-1	30%-60%
Ester	141-78-6	≤10%
Ketone	67-64-1	≤10%

**4. FIRST AID MEASURES**

Inhalation	: Move to fresh air. If not breathing, give artificial respiration. If unconscious, place in recovery position and seek medical advice immediately.
Skin Contact	: Remove contaminated clothing and wash affected area with plenty of soap and water. Get medical attention if irritation develops or persists.
Eye Contact	: Rinse cautiously with water for several minutes by lifting the eyelids to ensure complete flushing. Remove contact lenses, if present and easy to do. Get medical advice/attention if irritation persists.
Ingestion	: Rinse mouth with water. Never give anything by mouth to an unconscious person. DO NOT induce vomiting. IF vomit spontaneously, incline the body to avoid aspiration into respiratory tract. Get medical attention immediately.
Most Important Symptoms/ Effects, Acute and Delayed	: No information available
Indication of immediate medical attention and special treatment needed, if necessary.	: Treat symptomatically. Symptoms may be delayed.

5. FIRE FIGHTING MEASURES**Suitable Extinguishing Media**

Dry chemical, CO₂, Alcohol resistant foam, Fog, Sand or earth, Water spray.

Unsuitable Extinguishing Media

Water Jet

Specific hazards arising from the chemical.

Hazardous combustion products: Fumes, smoke, carbon dioxide, carbon monoxide and other toxic gases. May release vapour that form flammable mixtures at or above the flash point during combustion. Vapour formed may settle in low areas or can travel some distance along the ground or surface to ignition sources where they may ignite or explode.

Special protective equipment and precautions for fire-fighters.

Evacuate area. Remove all ignition sources. Stop leak if safe to do so. Stay upwind to avoid danger of vapours and hazardous decomposition products intoxication. Water spray/ fog may be used to disperse vapour and cool fire fighters and surrounding containers. Container may rupture from gas generation in a fire situation. Enter fire zone with adequate personnel protective equipment.

A self-contained breathing apparatus (SCBA) is recommended for indoor fires and any significant outdoor fires. For small outdoor fires, which may easily be extinguished with a portable fire extinguisher, use of an SCBA is optional. Respiratory and eye protection required for firefighting personnel.

6. ACCIDENTAL RELEASE MEASURES**Personal Precautions, Protective Equipment and Emergency Procedures:**

Keep people away from and upwind of spill/leak. Product is highly flammable volatile liquid and may form invisible vapour. Remove all ignition sources. Avoid breathing vapours and contact with the product. Vapour may accumulate to form explosive concentrations. Ensure adequate ventilation. Emergency personnel are recommended to wear appropriate/ adequate personal protective equipment including protective gloves/ protective clothing/ eye protection/ face protection.

Environmental Precautions:

Shut off leaks, if possible without personal risks. Prevent leaks from entering drains, sewer or watercourses. If the product contaminates rivers or any other water streams, inform respective authorities.

Methods and Material for Containment and Clean Up:

Contain spilled liquid with sand, earth or chemical absorbent. DO NOT use combustible absorbent such as sawdust. Dispose recover materials in accordance to local regulation.

7. HANDLING AND STORAGE
Precautions for safe handling:

Avoid inhalation of vapours. Use only non-sparking tools. To prevent fire caused by electrostatic discharge, equipment on all metal parts should be grounded. Use explosion proof equipment. Handling is performed in a well-ventilated place. Wear suitable protective equipment. Avoid contact with skin and eyes. Keep away from heat/sparks/open flames/ hot surfaces. Take precautionary measures against static discharges.

Conditions for safe storage, including any incompatibilities:

Storage condition: Ambient.

Store in a cool, dry, well-ventilated area. Keep away from heat, sparks, flame, ignition sources and incompatible materials. Inspect periodically for leakage.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION
Engineering Control Measures/Ventilation:

Use local exhaust ventilation to control the concentration below the exposure limit. Provide mechanical ventilation of confined spaces. Use explosion-proof ventilation equipment.

Occupational Exposure Limits (OEL) Values:

Guideline: 100ppm (TWA)

Personal Protection:


Respirator. Gloves. Goggles. Full suite. Footwear.

Respiratory Protection

In inadequately ventilated areas, where workplace limits are exceeded or when unpleasant odours exist, use approved respiratory protection.

Eye Protection

Use chemical goggles with side shields. If vapour exposure causes eye discomfort, use a full-face, supplied-air respirator to prevent other exposure.

Skin Protection

For repeated or prolonged contact, wear appropriate protective clothing, apron, and impervious gloves.



9. PHYSICAL AND CHEMICAL PROPERTIES	
Physical State: liquid	Form/Color: Clear colourless liquid
Odor: Aromatic Odour	Odor Threshold: N/A
pH: N/A	Freezing/Melting Point: -98°C
Boiling Point Range: 56- 145°C	Flash Point (TCC ASTM D56): -17°C
Flammability Limits: 1.2 -50vol%	Vapor Pressure: N/A
Auto-ignition Temperature: 385 °C	Vapour Density (101.3kPa/air = 1): >1.00
Specific Gravity @ 25°C: 0.837	Solubility in water (20 °C): Low solubility

10. STABILITY AND REACTIVITY	
Reactivity: Not reactive mixture	Chemical Stability: Stable under normal condition of use.
Possibility of Hazardous Reaction: Hazardous polymerization will not occur. Reacts violently with strong oxidants causing fire and explosion hazard.	Hazardous Decomposition Products: None known
Conditions to Avoid: Avoid any source of ignition. Avoid contact with heat, sparks, open flames and static discharge.	Incompatible Materials: Strong oxidising agents, acids and bases.

11. TOXICOLOGICAL INFORMATION	
Acute toxicity estimate:	
Acute oral toxicity	: LD50: 216mg/kg
Acute inhalation toxicity	: LC50: 5.31mg/l
Acute dermal toxicity	: LD50: 300mg/kg
Skin Corrosion/irritation	: Causes skin irritation and may cause skin dermatitis.
Serious eye damage/irritation	: Causes eye irritation.
Respiratory or skin sensitization	: No data available.
Germ cell mutagenicity	: No data available
Carcinogenicity	: Suspected of causing cancer.
Toxic to reproduction	: Cause adverse foetal developmental effects
Aspiration toxicity	: May cause mild to severe pulmonary injury and possibly death.
STOT - single exposure	: Causes CNS, eye and respiratory tract effect. May cause headache, dizziness, anaesthesia, drowsiness.
STOT - repeated exposure	: May cause brain & nervous system damage. May cause damage to kidney and liver through prolonged and repeated exposure.

**12. ECOLOGICAL INFORMATION****Acute Toxicity:**

Fish : LC/EC/IC50 > 1mg/l

Daphnia & other aquatic invertebrate : LC/EC/IC50 > 1mg/l

Algae : LC/EC/IC50 > 1mg/l

Mobility: Float on water. If product enters soil, it will be highly mobile and may contaminate groundwater.**Persistence/degradability:** Readily biodegradable.**Bioaccumulation:** Does not bioaccumulate significantly.**Other adverse Effects:** No data available.**13. DISPOSAL CONSIDERATIONS****Disposal methods**

At the time of disposal, users must re-evaluate the product disposal classification since the product had been used, transformed, mixed or contaminated.

Dispose of only in accordance with local, state, and federal regulations. Do not contaminate any lakes, streams, ponds, groundwater or soil.

Empty containers.

Empty containers may contain product residues (liquid and/or vapour) and can be dangerous. Do not pressurize, cut, weld, braze, solder, drill, grind or expose such containers to heat, flame, sparks, static electricity, or other sources of ignition; it may explode and cause injury or death. Empty drums should be completely drained, rinse thoroughly, properly banded and dispose in accordance to the law.

14. TRANSPORT INFORMATION

UN proper shipping name: F2 GP Thinner

UN number: 1993

Hazchem code: 3YE

Transport hazard class: 3, Flammable Liquid

Packing group: II

Environmental hazards: Not marine pollutant

15. REGULATORY INFORMATION**Safety, health, and environmental regulations specific for the hazardous chemical**

- Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013.
- Occupational Safety and Health (Use and Standards of Exposure of Chemicals Hazardous to Health) Regulations 2000.
- Environmental Quality Act 1974, Environmental Quality (Scheduled Wastes) Regulations 2005

16. OTHER INFORMATION

This information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process. Such information is to the best of LSC company's knowledge and believed accurate and reliable as of the date indicated. However, no representation, warranty or guarantee is made as to its accuracy, reliability or completeness. It is the user's responsibility to justify the suitability and completeness of such information for her particular usage.

Preparation Date: 1st July 2021**Revision Number:** 0**References** : SDS of Material Manufacturers